

Detecting Adverse Drug Reactions in Scientific Texts

Official recommendations surrounding adverse drug reactions (ADRs) are largely curated from independent reports or electronic health records (EHRs) submitted by drug manufacturers and clinicians. Such reports are time-consuming and tedious to label as they require technical expertise, thus ADRs are often under-reported and inefficient to study. Building an natural language model that is able to automatically crawl through scientific texts and extract information related to ADRs can help expedite medical intervention and development of treatments related to such conditions.

Significant work has been done on detecting ADRs in scientific texts. Burgt et al. developed a text mining algorithm to identify ADRs in Dutch EHRs (Burgt et al., 2024), combining R-scripts and public databases such as Medical Dictionary (MEDRA) for Regulatory Activities and Systematized Nomenclature of Medicine Clinical Terms (NOMED-CT) to determine related terms to find. Their algorithm showed a sensitivity of 73% and a positive predictive value of 70% in identifying ADRs. Other researchers such as Zhang et al. have looked at extracting ADRs on social media (Zhang et al., 2020), in an attempt to boost under-reported ADRs and expand the types of sources we can use to obtain information on them. Integrating shallow and deep linguistic features to train models, they found that their novel approach outperformed other state-of-the-art methods with an AUC of 94.44% and 88.97% on two datasets.

This project will make use of the data provided in the `ade_corpus_v2` benchmark, generating training, validation and testing datasets. Feature engineering such as building a dictionary of official ADR technical terms or a dictionary of sentiment terms to determine if texts related to ADR terms are adverse or positive might also be useful in facilitating useful pattern learning during training. Additionally, further research will be done to determine the most suitable type of model to be used for ADR detection tasks. Besides these, techniques such as cross-fold validation and adopting dropouts will also be incorporated during the training phase to reduce overfitting and improve model performance on detecting ADRs on unseen data. The final model shall be evaluated on the test set provided using standard model evaluation metrics such as precision, recall, f1, AUCs to determine model performance.

References

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